

Incumbency Protection as a Redistricting Criterion: Legal Analysis and Algorithmic Implications

Giovanni Della-Libera

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Abstract

Incumbency protection is one of the most commonly cited justifications for departures from neutral redistricting criteria, yet its legal status in federal and state law is poorly understood by practitioners on both sides of redistricting disputes. This paper provides a comprehensive legal analysis of incumbency protection as a redistricting criterion: its permissibility under federal constitutional doctrine, its interaction with the Voting Rights Act of 1965, and the state constitutional frameworks that govern its use. We identify three federal cases that define the constitutional parameters: *Karcher v. Daggett* (1983), which identifies incumbency protection as a legitimate objective that may justify minor population deviations; *Cox v. Larios* (2004), which limits that permission to deviations not exceeding constitutional thresholds; and *Thornburg v. Gingles* (1986), which creates a tension between incumbency protection and VRA §2 compliance in majority-minority districts. We then analyse the implications of this doctrine for the BISECT algorithmic redistricting system, which neither protects nor targets incumbents. Our central legal conclusion is that BISECT’s constitutional silence on incumbency—the algorithm receives no incumbent address data and produces incumbency outcomes purely as geometric consequences of population and compactness optimisation—is legally sound under all three levels of the analysis: federal constitutional doctrine does not require incumbent protection; VRA compliance does not require incumbent addresses; and state constitutional frameworks that permit incumbency consideration do not mandate it. The algorithmic map’s incumbency outcomes (documented in I.1–I.3) are indistinguishable from what a neutral commission instructed to ignore incumbency would produce, establishing BISECT as a constitutionally defensible redistricting methodology on incumbency grounds.

Keywords

Incumbency Protection, Redistricting Criteria, *Karcher v. Daggett*, *Burdick v. Takushi*, *Thornburg v. Gingles*, Voting Rights Act, Algorithmic Redistricting, Constitutional Compliance, BISECT, State Constitutional Law

1 Introduction: The Incumbency Objection to Algorithmic Redistricting

When a state legislature or redistricting commission proposes an algorithmic map for legislative approval or court review, incumbents and their allies reliably raise a specific objection: the algorithm does not consider incumbency. This objection takes several forms. In legislative proceedings, it appears as a policy argument—“We have a responsibility to maintain continuity in representation”—invoking the constituents’ interests in an uninterrupted relationship with their existing representative. In litigation, it appears as a legal argument—“The enacted map appropriately considered the legitimate redistricting criterion of preserving the cores of prior districts and avoiding contests between incumbents”—citing the Supreme Court’s language in *Karcher v. Daggett*¹ as authority for the proposition that the algorithm’s failure to consider incumbency is a legal defect.

Both forms of the objection deserve a thorough response. This paper provides that response, grounded in the federal constitutional doctrine, VRA requirements, and state constitutional law that govern redistricting in the three states studied in I.1–I.3.

1.1 The Objection in Three Forms

The incumbency objection to algorithmic redistricting presents itself in three distinct legal framings:

The *Karcher* objection. The Supreme Court in *Karcher v. Daggett* identified “avoiding contests between incumbents” as a “legitimate objective” that may justify minor population deviations from mathematical equality. Opponents of algorithmic maps argue from this language that incumbency consideration is a constitutionally endorsed redistricting criterion, and that a redistricting methodology that ignores it is legally incomplete or defective.

¹*Karcher v. Daggett*, 462 U.S. 725, 740 (1983).

The VRA objection. In states with majority-minority districts, protecting minority incumbents is sometimes argued to be required by VRA §2 as part of the remedy for vote dilution. If BISECT maps pair a minority incumbent with an Anglo incumbent, opponents argue that the algorithmic indifference to incumbency has produced a map that dilutes minority voting strength by forcing a primary contest that the minority incumbent may lose.

The state law objection. Several states have redistricting criteria statutes or commission guidelines that explicitly list “preserving communities of interest” or “maintaining incumbency” as permissible considerations. Opponents of algorithmic maps in these states argue that the algorithm’s failure to consider these factors violates state statutory law.

1.2 The Response in Three Parts

This paper develops a three-part response:

1. Incumbency protection is permissible under federal constitutional doctrine, but not required. *Karcher*’s identification of it as a “legitimate objective” does not mandate its consideration; it merely confirms that a mapmaker who considers it is not constitutionally defective. An algorithm that does not consider it is equally not constitutionally defective.
2. The VRA objection conflates the VRA’s requirements for district drawing with requirements for *how* districts are drawn. The VRA requires that minority voters have an equal opportunity to elect candidates of their choice; it does not require that the redistricting process protect any specific incumbent’s electoral prospects. When BISECT maps produce VRA-compliant majority-minority districts, the VRA is satisfied regardless of where specific incumbents’ homes happen to fall.
3. State redistricting criteria that permit incumbency consideration impose a duty to consider, not a mandate to protect. An algorithmic process that systematically excludes incumbency data cannot “consider” incumbency in the way state statutes contemplate; but the proper remedy—if one is needed—is to run the algorithm with an awareness of its incumbency outputs and to make post hoc adjustments where VRA compliance requires them, not to inject incumbency data into the algorithmic optimisation.

1.3 Paper Organisation

Section 2 reviews the federal constitutional doctrine on incumbency, covering the full lineage from *Karcher* through *Cox v. Larios*. Section 3 develops the *Karcher* doctrine in detail,

clarifying what the “legitimate objective” language does and does not authorise. Section 4 analyses the VRA interaction, drawing on *Thornburg v. Gingles* and its progeny. Section 5 surveys state constitutional frameworks and develops the algorithmic indifference standard. Section 6 concludes with practical guidance for practitioners using BISECT in litigation.

2 Incumbency Protection as a Permissible Redistricting Criterion: Federal Constitutional Doctrine

2.1 The Historical Development of Incumbency as a Redistricting Criterion

The use of incumbent home addresses in redistricting predates modern civil rights law. Reconstruction-era state legislatures routinely drew districts to protect sitting members, and this practice was so thoroughly embedded in the redistricting tradition that early federal reapportionment litigation rarely addressed it [Backstrom et al., 1978]. The Supreme Court’s entry into redistricting in *Baker v. Carr*² and *Reynolds v. Sims*³ established the one-person, one-vote principle but said nothing about whether mapmakers could consider incumbent addresses in drawing equal-population districts.

The Court’s first significant engagement with incumbency as a redistricting criterion came in *Karcher v. Daggett* (1983), where New Jersey’s partisan gerrymander was challenged on equal population grounds. Justice Brennan’s majority opinion, striking down the New Jersey map, noted that minor population deviations might be justified by “legitimate objectives,” including “making districts compact, respecting municipal boundaries, preserving the cores of prior districts, and avoiding contests between incumbents.” This list was explicitly non-exhaustive; the Court was cataloguing justifications that had historically been accepted, not establishing a definitive taxonomy of permissible criteria.

2.2 Permissive vs. Mandatory: The Doctrinal Distinction

The crucial doctrinal distinction that the incumbency objection typically elides is between *permissible* criteria (which a mapmaker may consider) and *mandatory* criteria (which a mapmaker must consider). The Supreme Court has been consistent in characterising incumbency protection as permissible. It has never held that any redistricting body is constitutionally required to consider incumbent home addresses or to draw districts that place each incumbent

²*Baker v. Carr*, 369 U.S. 186 (1962).

³*Reynolds v. Sims*, 377 U.S. 533 (1964).

in a protected home seat.

This distinction was clarified in *Cox v. Larios*⁴ where the Court affirmed that incumbency protection, while legitimate, cannot justify population deviations that would otherwise fail constitutional scrutiny. The Court’s reasoning implicitly confirms the permissive character of the criterion: a mandatory criterion would be a reason to accept whatever population deviation it required, but a permissible criterion is only a potential justification for deviations that are otherwise within constitutional tolerance.

2.3 *Burdick v. Takushi* and Election Administration Stability

*Burdick v. Takushi*⁵ addressed Hawaii’s prohibition on write-in voting. Justice White’s majority opinion established the sliding-scale balancing test for election law challenges: courts weigh the state’s asserted interest against the burden on First and Fourteenth Amendment rights, with more severe restrictions requiring stronger justifications. Justice White identified “election administration” stability as a cognisable state interest, and subsequent lower courts have cited this language to support the proposition that incumbent protection—which promotes continuity in representation and reduces the disruption of primary contests between established officeholders—serves a state interest in election administration stability [Pildes, 2004].

The application of *Burdick* to redistricting is strained. *Burdick* addressed a ballot restriction affecting individual voters, not a legislative line-drawing choice. But the lower court adoption of *Burdick*’s interest-balancing framework for redistricting incumbency questions means that practitioners must address it. The response is straightforward: even accepting the *Burdick* framework, the state’s interest in election administration stability does not require that every redistricting plan protect every incumbent. The interest is served by having some continuity in representation, not by guaranteeing every sitting member a safe home district.

2.4 Summary of Federal Doctrine

Table 1 summarises the federal case law on incumbency as a redistricting criterion.

⁴*Cox v. Larios*, 542 U.S. 947 (2004), summarily affirming 300 F.Supp.2d 1320 (N.D. Ga. 2004).

⁵*Burdick v. Takushi*, 504 U.S. 428 (1992).

Table 1: Federal case law on incumbency as a redistricting criterion.

Case	Year	Holding on Incumbency
<i>Karcher v. Daggett</i>	1983	Avoiding incumbent contests is a “legitimate objective” that may justify minor pop. deviations; not mandatory
<i>Burdick v. Takushi</i>	1992	Election administration stability is a cognisable state interest; indirect support for incumbency consideration
<i>Cox v. Larios</i>	2004	Incumbency protection cannot justify pop. deviations exceeding constitutional tolerance; confirms permissive (not mandatory) character
<i>Rucho v. Common Cause</i>	2019	Partisan gerrymandering non-justiciable federally; state courts may evaluate incumbency as tool of partisan advantage

Federal doctrine is consistently permissive: incumbency may be considered, not that it must be. No federal court has required a redistricting plan to protect any specific incumbent.

3 *Karcher v. Daggett* and Population Deviations

3.1 The *Karcher* Doctrine in Detail

Justice Brennan’s majority opinion in *Karcher v. Daggett* is the locus classicus for incumbency as a redistricting criterion, and it deserves careful attention because its language is frequently misread in redistricting proceedings.

The holding of *Karcher* was that New Jersey’s congressional map, which contained population deviations of 0.6984%, was unconstitutional because the state failed to demonstrate that those deviations were necessary to achieve a legitimate state objective. Justice Brennan’s majority accepted that the deviations were small in absolute terms but held that the Equal Population principle demanded justification for any deviation from mathematical equality in congressional apportionment [U.S. Supreme Court, 1983].

The language on legitimate objectives appeared in the context of explaining what kinds of justifications would suffice to uphold a map with minor deviations:

“Any number of consistently applied legislative policies might justify some variance, including, for example, making districts compact, respecting municipal boundaries, preserving the cores of prior districts, and avoiding contests between incumbents.”

Three features of this passage are critical for the incumbency objection. First, the passage is framed hypothetically: it describes what “might” justify deviations, not what must be done. Second, it is explicitly a non-exhaustive list (“for example”), indicating that the Court was not enumerating mandatory criteria. Third, the passage appears in a section explaining why the New Jersey map *failed* the test—the state did not demonstrate that the listed or other objectives justified its deviations. The Court was not endorsing the criteria; it was explaining what a successful justification would need to show.

The misreading of *Karcher* in incumbency litigation typically proceeds as follows: the *Karcher* dicta identifies incumbency protection as a legitimate objective; a legitimate objective is a required consideration; therefore, any redistricting plan that does not consider incumbency fails the *Karcher* test. Each step in this syllogism is incorrect. “Legitimate objective” means an objective that is not constitutionally prohibited, not one that is constitutionally required. Redistricting bodies are free to pursue or not pursue any legitimate objective, so long as they satisfy the constitutional requirements of equal population, non-dilution of minority voting strength, and contiguity.

3.2 The *Cox v. Larios* Limitation

Cox v. Larios provides the clearest statement of the limits of incumbency as a justification for population deviations. The three-judge district court in the Northern District of Georgia found that Georgia’s state legislative maps contained population deviations of up to 9.98% in the state Senate and 4.98% in the state House, justified in part by the legislature’s desire to protect incumbent legislators. The court held that using incumbency protection as a primary justification for deviations of this magnitude was unconstitutional: the deviations were too large to be justified by any legitimate interest, including incumbency protection [U.S. Supreme Court, 2004].

The Supreme Court’s summary affirmance of *Cox v. Larios* confirmed that the district court’s analysis was correct, at least as applied to the facts. The holding can be read as establishing a rough proportionality constraint: small deviations may be justified by legitimate interests including incumbency; large deviations cannot. For congressional redistricting, where any deviation from mathematical equality must be justified [U.S. Supreme Court, 1983], the *Cox* holding implies that incumbency protection cannot justify any deviation that

is not the *necessary* result of pursuing the incumbent-protection objective—a high standard that enacted maps frequently fail to meet.

3.3 Algorithmic Redistricting and Population Deviation

The BISECT algorithm produces congressional maps that satisfy the equal-population constraint without any incumbency-related deviation. By operating on census tracts rather than precincts, and by explicitly balancing population at each bisection step, the algorithm produces maps where the maximum population deviation across districts is well below 1% in all three states studied. This performance is not dependent on incumbency protection—the algorithm achieves near-perfect population equality purely through its population-balance optimisation.

This means that the *Karcher/Cox* analysis is largely irrelevant to algorithmic redistricting: the algorithm does not introduce population deviations that require justification by incumbency or any other criterion. The *Karcher* line of cases addresses the justification for deviations that the algorithm does not produce. The incumbency objection, framed as a *Karcher* challenge, therefore fails at the threshold: there are no deviations to justify.

3.4 Three Uses of Algorithmic Incumbency Analysis in Litigation

The I.1–I.3 empirical results can be deployed in redistricting litigation in three distinct ways.

Establishing non-randomness of enacted pairing rates. I.1’s binomial test demonstrates that enacted map pairing rates are statistically inconsistent with random redistricting. This is evidence that mapmakers actively used incumbent address data to minimise pairings—a redistricting choice, not a geographic necessity. In litigation challenging the enacted map, this evidence supports the inference that the map was drawn with at least some degree of strategic incumbency protection.

Providing a neutral reference distribution. The BISECT map provides a neutral reference point for all three incumbency metrics: pairing rate, safe-seat fraction, and open-seat count. Courts can use this reference distribution to assess whether the enacted map’s incumbency outcomes are within the range of what geographic redistricting would produce, or whether they represent deliberate departures from geographic chance. A large positive incumbency protection premium (as documented in I.0–I.3) supports the inference of strategic mapmaking.

Quantifying the intentional incumbency protection premium. The IPP metric (I.1) provides a single number that quantifies the departure from geographic neutrality on incumbent pairing. Courts evaluating whether an enacted map’s use of incumbency was proportionate to its legitimate objectives can use the IPP to assess the degree of the departure: a large IPP (enacted pairing rate far below the geographic baseline) suggests that incumbency was not merely considered but was a dominant criterion that displaced other redistricting objectives.

4 VRA Interaction: When Incumbency and Minority Representation Conflict

4.1 VRA §2 and Incumbency: The *Gingles* Framework

The Voting Rights Act of 1965, §2, as interpreted in *Thornburg v. Gingles*⁶ and its progeny, prohibits redistricting plans that result in the denial or abridgement of the right to vote on account of race or color. The *Gingles* three-part test for demonstrating vote dilution requires plaintiffs to show: (1) the minority group is sufficiently large and geographically compact to form a majority in a single-member district; (2) the minority group is politically cohesive; and (3) the white majority votes sufficiently as a bloc to usually defeat the minority-preferred candidate.

When all three prongs are met, the redistricting body must draw a majority-minority district in the relevant area, unless doing so would require deviation from other constitutionally or statutorily required criteria [U.S. Supreme Court, 1986]. The question for algorithmic redistricting is whether the VRA’s majority-minority district requirement independently mandates that the district be drawn in a way that protects the incumbent minority representative’s home address.

4.2 The VRA Does Not Require Incumbency Protection

The answer is no. The VRA requires a district in which minority voters have an equal opportunity to elect candidates of their choice—not that the specific candidate currently representing that district be protected from competitive challenges. *Gingles* and its progeny focus on the electoral opportunity available to minority voters as a group, not on the electoral security of any individual incumbent.

⁶*Thornburg v. Gingles*, 478 U.S. 30 (1986).

This distinction is practically significant because minority incumbents, like all incumbents, benefit from incumbency advantage that provides electoral security even without gerrymandered district protection. A minority incumbent who faces a primary opponent following redistricting is not, by that fact alone, the victim of a VRA violation; the VRA violation would require showing that the redistricting has diminished the minority group’s opportunity to elect their preferred candidate below the level that existed prior to redistricting.

4.3 The Tension When Incumbency and VRA Conflict

A genuine tension arises when protecting a minority incumbent’s residency core conflicts with the minority community’s VRA-required opportunity district. This can occur in two scenarios.

Scenario 1: Anglo incumbent in majority-minority area. If an Anglo incumbent represents a district that has, through demographic change, become majority-minority, the VRA may require redrawing the district to create a majority-minority district. Protecting the Anglo incumbent’s home tract may conflict with the VRA requirement to create the majority-minority district. Courts have consistently held that the VRA requirement takes precedence over incumbent protection in this scenario [Karlan, 1995].

Scenario 2: Minority incumbent in a contested area. If a minority incumbent represents a district that is being split or reconfigured, the VRA may appear to require drawing boundaries that protect the incumbent’s district as a majority-minority seat. However, this scenario is more complex: the VRA requires a majority-minority *district*, not a majority-minority district that includes a specific incumbent’s home address. The algorithmic map may produce a majority-minority district that satisfies the *Gingles* test without including the specific incumbent’s home tract.

4.4 Algorithmic Maps and VRA Compliance

The BISECT algorithm does not automatically produce VRA-compliant maps; VRA compliance is evaluated *ex post* and, where required, the algorithm can be re-run with VRA constraints (see the `--weights-override vra-aligned` flag in the BISECT CLI documentation). The key point for the incumbency analysis is that VRA compliance and incumbency protection are analytically separable:

1. A BISECT map that produces a majority-minority district in the geographically appropriate area satisfies VRA §2 for that area, regardless of whether the minority incumbent’s home tract falls within that district.
2. If the minority incumbent’s home tract does not fall in the majority-minority district, the incumbent may seek to run in that district (residency requirements for Congress are at the state level; most states require only that a candidate reside in the state, not the district), or may seek election in an adjacent district, or may retire. None of these outcomes is a VRA violation.
3. If the majority-minority district required by the VRA happens to not include the existing minority incumbent’s home tract, the appropriate resolution is to adjust the map’s boundaries to include the incumbent’s tract within the majority-minority district—a minor adjustment that does not require re-running the full algorithm—rather than to inject incumbency data into the algorithmic optimisation.

4.5 State Constitutional Constraints

Survey of state constitutional provisions reveals a spectrum of approaches to incumbency as a redistricting criterion.

Explicit permission. Several states explicitly permit consideration of incumbency in their redistricting criteria statutes or constitutional provisions. California’s commission guidelines permit “maintaining communities of interest and neighbourhoods” as a criterion, which has been interpreted to include maintaining existing constituent-representative relationships [National Conference of State Legislatures, 2021]. Ohio’s constitutional redistricting criteria (pre-amendment) allowed “maintaining the cores of prior districts,” which provided an implicit basis for incumbency protection.

Explicit prohibition. Arizona and California’s independent redistricting commission enabling statutes explicitly prohibit the commission from drawing districts to favour any incumbent or political party. This prohibition is directional (commissions cannot protect incumbents) but does not speak to whether algorithms that are indifferent to incumbency comply with the prohibition. We argue that they do: an algorithm that neither protects nor targets incumbents satisfies a prohibition on incumbency protection.

Silent frameworks. Most state constitutions and redistricting statutes do not mention incumbency at all, leaving the question to the redistricting body’s discretion within the

bounds of constitutional and statutory criteria. In these states, the algorithmic indifference to incumbency is legally unremarkable: the algorithm is not required to consider incumbency, and its failure to do so cannot be characterised as a legal defect.

5 State Constitutional Frameworks and the Algorithmic Indifference Standard

5.1 The Algorithmic Indifference Standard

We propose the following standard for evaluating whether an algorithmic redistricting methodology is constitutionally sound on incumbency grounds:

Definition 1 (Algorithmic Indifference Standard). *An algorithmic redistricting methodology satisfies the algorithmic indifference standard on incumbency if and only if:*

1. *The algorithm receives no incumbency data (no incumbent home addresses, no prior district boundary data used for incumbency-protection purposes) as an input.*
2. *The algorithm’s output is determined solely by constitutionally required or permitted criteria (population balance, contiguity, compactness, community of interest) that do not include incumbency.*
3. *The algorithm’s incumbency outcomes are statistically consistent with what a random geographic redistricting process would produce given the same geographic constraints.*

The BISECT algorithm satisfies all three prongs. It receives no incumbent addresses. Its optimisation target is population balance and geometric compactness. And its incumbency outcomes (I.1–I.3) are statistically consistent with the geographic random baseline across NC, WI, and TX.

A methodology that satisfies the algorithmic indifference standard cannot be challenged on the ground that it “fails to consider” incumbency in any constitutionally meaningful sense: the constitutional doctrine permits but does not require incumbency consideration, and the algorithm’s outputs are equivalent to what a truly neutral process would produce.

5.2 Post-*Rucho* State Court Applications

The algorithmic indifference standard has particular relevance in state court proceedings following *Rucho v. Common Cause*⁷ State courts evaluating redistricting plans under state

⁷*Rucho v. Common Cause*, 588 U.S. 684 (2019).

constitutional free-elections, equal protection, or partisan fairness claims have increasingly adopted the approach of comparing enacted maps to neutral algorithmic baselines. The North Carolina Supreme Court in *Harper v. Hall*⁸ and the Pennsylvania Supreme Court in *League of Women Voters v. Commonwealth*⁹ both used neutral map comparisons to evaluate whether enacted maps departed from what neutral redistricting would produce.

In these proceedings, incumbents may argue that the algorithmic baseline is unfair because it does not protect incumbency. The algorithmic indifference standard provides the response: an algorithm that is indifferent to incumbency produces outcomes that are statistically equivalent to what neutral redistricting—redistricting that is directed to achieve only constitutionally required criteria—would produce. The algorithm’s outcomes are, by construction, the outputs of a fair process that applies no thumb on the scale for or against any specific incumbent.

5.3 The Expert Witness Application

For expert witnesses proposing BISECT maps in redistricting litigation, the legal analysis in this paper supports three specific arguments.

Incumbency consideration is not required. The first and most direct argument: federal constitutional doctrine does not require any redistricting methodology to consider incumbency. *Karcher*’s identification of incumbency as a legitimate objective means that mapmakers may consider it; it does not mean that methodologies that do not consider it are constitutionally defective. An algorithm that ignores incumbency is no different in constitutional status from a commission that is instructed not to consider incumbent addresses in drawing its map— a perfectly lawful instruction that courts have endorsed in Arizona’s and California’s commission systems.

Algorithmic outcomes are equivalent to neutral redistricting. The second argument draws on I.1–I.3: the BISECT algorithm’s incumbency outcomes are statistically indistinguishable from what a random geographic redistricting process would produce. This means that the algorithm is not more disruptive to incumbency than any other neutral redistricting process. An expert witness can testify that the algorithmic map’s pairing rate, safe-seat fraction, and open-seat count are all within the expected range for a geographically constrained redistricting process with no incumbency input—the algorithm is neutral on incumbency because its outputs reflect geographic chance, not strategic choice.

⁸*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

⁹*League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

The enacted map’s incumbency outcomes are non-random. The third argument is the flip side: I.1’s binomial tests demonstrate that the enacted map’s incumbency outcomes—zero pairings in NC and WI, significantly below-baseline pairing in TX—are inconsistent with random redistricting. This is circumstantial evidence that the enacted mapmakers used incumbent address data to engineer protective districts. The contrast between algorithmic neutrality and enacted strategic protection provides courts with a precise quantitative measure of the incumbency manipulation that the enacted map embeds.

5.4 Practitioner Decision Table

Table 2 summarises the legal analysis for practitioners evaluating whether to deploy BISECT maps in redistricting proceedings.

Table 2: Decision table for incumbency objections to algorithmic redistricting.

Objection	Legal Standard	Response
Algorithm ignores incumbency	<i>Karcher</i> : permissive, not mandatory	No court requires incumbency consideration; algorithm satisfies all required criteria
Algorithm produces pairing	No constitutional prohibition on pairing	Pairing rate is consistent with geographic baseline; enacted zero-pairing requires incumbency data
Algorithm produces unsafe seats	VRA requires minority opportunity, not safe seats	VRA compliance is evaluated ex post; majority-minority districts preserved where required
State statute permits incumbency	Permit $\neq$ require	Algorithm is not prohibited from ignoring incumbency; no state statute mandates protection
Algorithm disrupts continuity	<i>Burdick</i> : election admin interest	Algorithm disruption is equivalent to neutral redistricting; more continuity than random

6 Conclusion

This paper has provided a comprehensive legal analysis of incumbency protection as a redistricting criterion and its implications for algorithmic redistricting. The analysis rests on three pillars of federal constitutional doctrine: *Karcher v. Daggett*’s identification of incumbency protection as a permissible but not mandatory objective; *Cox v. Larios*’s limitation of incumbency protection to contexts where it justifies only minor population deviations; and *Thornburg v. Gingles*’ establishment of VRA §2 requirements that are independent of and take precedence over incumbency protection in majority-minority district contexts.

From these doctrinal foundations, we derived the algorithmic indifference standard: a redistricting methodology that receives no incumbency data, optimises only constitutionally required or permitted criteria, and produces incumbency outcomes consistent with the geographic random baseline is constitutionally sound under federal law. The BISECT algorithm satisfies this standard, and the empirical evidence in I.1–I.3 confirms that its incumbency outcomes are statistically indistinguishable from what neutral redistricting would produce.

The state constitutional analysis reveals a spectrum of approaches to incumbency as a redistricting criterion, ranging from explicit prohibition (Arizona, California commission systems) to explicit permission (California commission guidelines, Ohio pre-amendment criteria) to silence (most state constitutions). In all three categories, the algorithmic indifference standard provides a defensible position: the algorithm is neither prohibited by the explicit-prohibition states (because it neither protects nor targets incumbents) nor required to protect incumbents by the explicit-permission states (because permission is not mandate) nor constrained by the silent states (because no constraint is stated).

For practitioners deploying BISECT maps in redistricting litigation, the central message is that the incumbency objection to algorithmic redistricting, while rhetorically effective, is legally unfounded. No federal constitutional provision requires incumbency protection; no VRA provision requires it; and no state constitutional or statutory provision mandates it in any state we have studied. The algorithm’s indifference to incumbency is not a defect—it is a design choice that produces incumbency outcomes equivalent to what a neutral geographic redistricting process would produce, and those outcomes are constitutionally sound.

The most powerful response to the incumbency objection in litigation is therefore empirical rather than doctrinal: the BISECT map’s incumbency outcomes are within the range of geographic chance, while the enacted map’s outcomes are significantly below the baseline, demonstrating that enacted incumbent protection requires strategic use of incumbency data that the algorithm does not access. This shifts the burden of justification onto the enacted map—the party defending the map must explain why the enacted map’s near-zero pairing

rate and above-baseline safe-seat fraction are a geographic necessity rather than a strategic choice.

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